

Generalizable Decision-Making for CAVs at Unsignalized Intersections

A Deep Reinforcement Learning Approach — Baseline Reproduction & Path to
Generalization

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CSE 4610: Design Project

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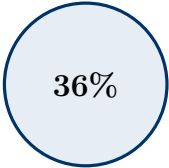
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Outline

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- 2 Related Work
- 3 Methodology
- 4 Implementation
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The Problem



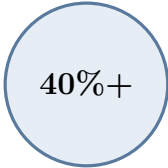
36%

of crashes are
intersection-related



\$518B

annual cost of in-
tersection crashes

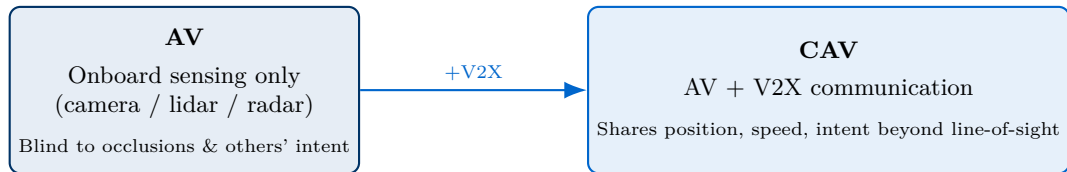


40%+

of collisions occur
at intersections ev-
ery year globally

- Traffic lights control at the lane-group level, not per vehicle
- Fixed timing → stopping even when the road is clear
- Unsignalized crossings: high risk from occlusion & uncertain intent
- No shared protocol → right-of-way must be inferred

Motivation: AV vs. CAV



Signal-Free AIM

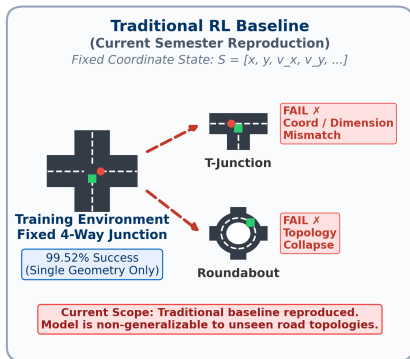
Individual, continuous negotiation replaces group-level stopping

Deep RL

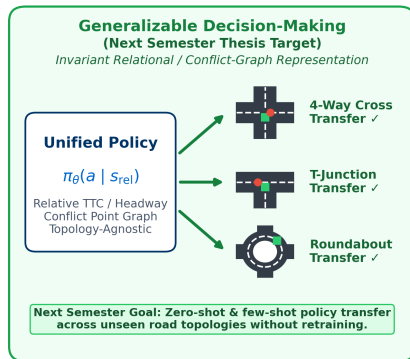
Learns adaptive go/yield behavior under stochastic traffic

Main Research Problem: Generalizable Decision-Making

Core Question: How can an autonomous vehicle make safe crossing decisions that **generalize across arbitrary, unseen intersection topologies** without retraining?



Current Semester (Completed): Reproduced traditional RL baseline (fixed 4-way); demonstrated that single-geometry models **do not generalize**.



Next Semester (Thesis Goal): Develop topology-invariant relational representations to achieve **zero-shot policy transfer** across unseen layouts.

Key Objectives

Safety

Near-zero collisions across randomized, dense turning maneuvers.

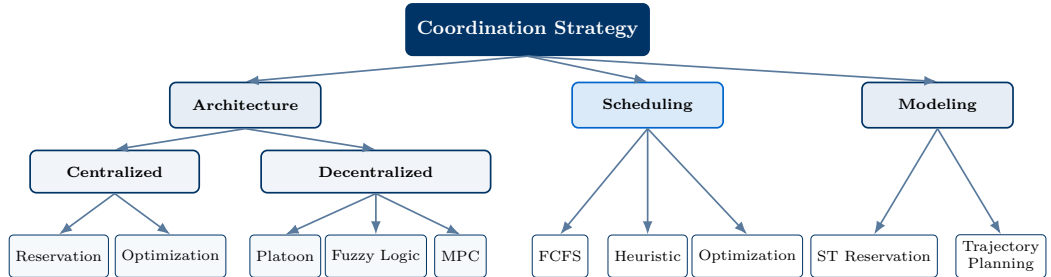
Efficiency

Maximize crossing speed and throughput without signal delays.

Generalizability

Zero-/few-shot policy transfer across unseen geometries (Next Sem.).

Taxonomy of Coordination Strategies



What the Literature Shows

Axis	Most popular	Runner-up
Goal	Safety + Efficiency (37%)	Safety alone (26%)
Architecture	V2V (159 papers)	V2I (119 papers)
Scheduling	Optimization (277 papers)	Heuristic (20 papers)
Modeling	Trajectory Planning	Spatio-Temporal Reservation

RL Literature Thread I: Single-Ego Decision Making

Paper	Idea	Architecture	Key finding
Isele et al. (2018)	DQN vs. TTC + occlusion “creeping”	Single-agent, no comm.	28% faster, <1% collisions
Wu et al. (2019)	Decentralized coordination Q-learning	Multi-agent, sparse comm.	Beats FCFS/LQF/Webster
Kai et al. (2020)	One policy, 3 maneuvers via task vector	Single-agent	Matches/beats 3 separate DQNs
Shu et al. (2022)	Weight transfer across maneuvers	Single-agent	Same-task 100%, cross-task >70%

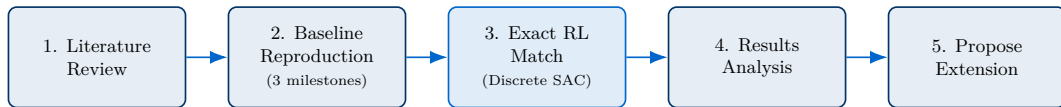
RL Literature Thread II: Graph Representations & Cooperative AIM

Paper	Contribution	Setting
Chen et al. (2025)	Scenario-agnostic “lane-gap” + PPO + CILQR	Single-ego, multi-scenario (93–97%)
Schier et al. (2023)	Vehicle+road graph, learned edges	Predicts right-of-way at 99.5% acc.
Klimke et al. (2022)	Edge-augmented RGCN + TD3, controls <i>all</i> vehicles	True cooperative AIM
Huamán Reyna et al. (2026)	Survey, 53 papers	Hybrid methods score best overall

Where we fit

- Single-ego decision-making, multi-task state (like Kai et al., Xiao et al.)
- Not centralized AIM (Klimke et al.) — tractable scope, still comparable

Methodology at a Glance



Each stage builds on the previous milestone's validated pipeline — detailed next.

“Decision-Making for AVs in Random Task Scenarios at Unsignalized Intersection Using Deep RL”

Three contributions

1. **Mix-Attention Network** (replaces plain MLP)
2. **Task-intent variable** μ_a
3. **Segmented replay buffer**

Setting

4-way, single-lane, unsignalized; random task/episode; **highway-env**; Discrete SAC, 50,000 episodes

Disclaimer: Scope of This Work

Reproduction & benchmarking study — not our primary contribution

Build a working pipeline, understand the environment deeply, then propose a novel direction.

Acknowledged limitations

- Original source code not released
- Some hyperparameters / reward constants only partially specified
- Reproduction is faithful, but not byte-for-byte

Research Progression: Three Milestones

M1 — Basic Ring Stabilization

Single-agent PPO, 230m ring, 21 IDM cars

Result: damped phantom shockwaves



M2 — Phase 1 Exploratory Reproduction

PPO, 4-way intersection, Original vs. Config B ($+\mu_a$)

Result: discovered left-turn vulnerability under heavy flow



M3 — Phase 2 Reproduction (Traditional Baseline)

Discrete SAC (twin critics, auto-tuned α), 50k episodes on fixed 4-way

Result: 99.52% success, 0% collisions — established baseline for generalization

State Space Formulation

Observation vector (48 m radar radius, up to 9 nearest surrounding vehicles):

$$\mathbf{S} = [S_{\text{ego}}, S_1, S_2, \dots, S_9]$$

Ego Vehicle State ($S_{\text{ego}} \in \mathbb{R}^6 \text{ or } 7$)

$$\mathbf{S}_{\text{ego}} = [p_{\text{ego}}, x, y, v_x, v_y, \phi, (\mu_a)]$$

- $p_{\text{ego}} \in \{0, 1\}$: **Existence flag** (1 = present)
- x, y : **Global position** coordinates (m)
- v_x, v_y : **Absolute velocity** along x, y (m/s)
- ϕ : **Heading angle** / orientation (rad)
- μ_a : **Task-intent feature** (*Config B only*)
Encodes route (left/straight/right) & exit progress

9 Surrounding Vehicles ($S_i \in \mathbb{R}^6$ each)

$$\mathbf{S}_i = [p_i, \Delta x_i, \Delta y_i, \Delta v_{x,i}, \Delta v_{y,i}, \phi_i]$$

- $p_i \in \{0, 1\}$: **Presence flag** (0 if empty / padded)
- $\Delta x_i, \Delta y_i$: **Relative position** ($x_i - x, y_i - y$) (m)
- $\Delta v_{x,i}, \Delta v_{y,i}$: **Relative velocity**
($v_{x,i} - v_x, v_{y,i} - v_y$) (m/s)
- ϕ_i : **Heading angle** of vehicle i (rad)
- **Ordering**: Sorted ascending by distance
 $d_i \leq 48$ m

Action Space & Reward Function

Action (discrete, 3 values)

$$A = \{\text{accel}, \text{idle}, \text{decel}\}$$

$$a = K_{P,A}(v_{tar} - v), \quad K_{P,A} = 5$$

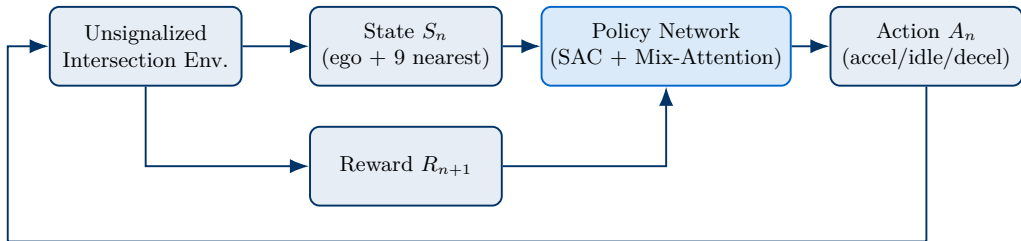
Lateral control is a fixed PD controller
— RL controls only longitudinal speed.

Reward

$$R = \begin{cases} +30 & \text{reach destination} \\ -120 & \text{collision} \\ +0.1 & \text{reach target speed} \end{cases}$$

- Sparse terminal terms (safety + completion)
- Small dense term rewards cruising
- Collision penalty $\gg$ arrival reward

System Pipeline



Discrete Soft Actor-Critic (Discrete SAC)

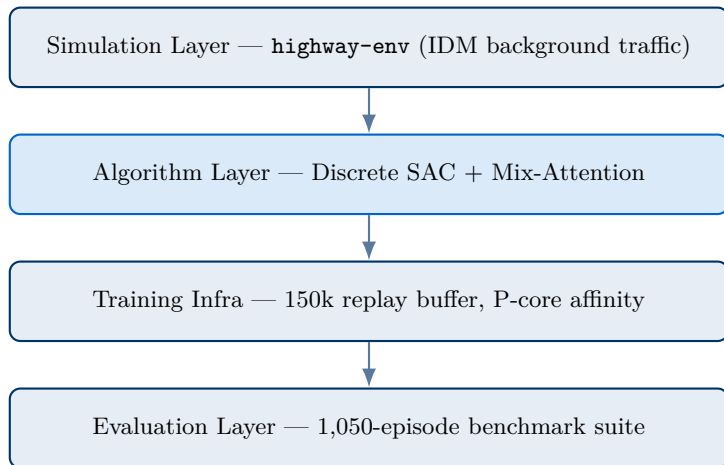
Why SAC over PPO

- Off-policy $\rightarrow$ sample efficient
- Max-entropy $\rightarrow$ broad exploration
- Auto-tuned α — no fragile hyperparameter

$$\pi^* = \arg \max_{\pi} \sum_n \mathbb{E}[\gamma^n (R_n + \alpha \mathcal{H}(\pi(\cdot|S_n)))]$$

- Twin Q-critics reduce overestimation bias
- Discrete policy: softmax over 3 actions

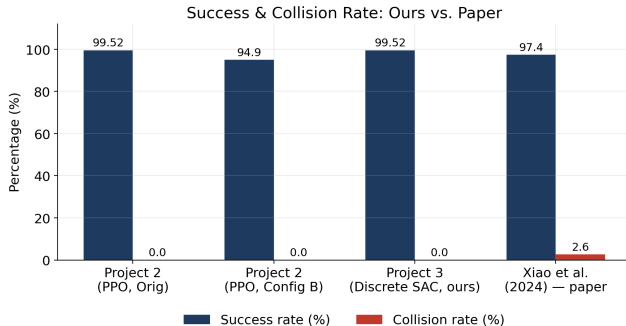
Implementation Stack



Simulation & Training Setup

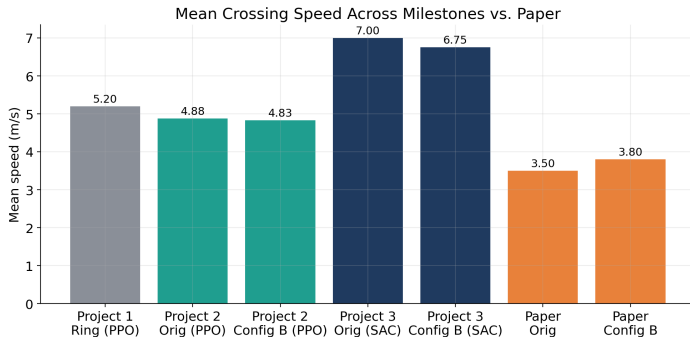
	Project 1: Ring	Project 2: Phase 1	Project 3: Phase 2	Paper
Scenario	230m ring	4-way	4-way	4-way
Sim. engine	Flow + SUMO	Flow + SUMO	highway-env	highway-env
Algorithm	PPO	PPO	Discrete SAC	Discrete SAC
Network	Actor-Critic MLP	[32, 32, 32]	[256, 256, 64, 3]	[256, 256, 64, 3]
Obs. space	3-dim POMDP	60 / 61-dim	60 / 61-dim	60 / 61-dim
Action space	Continuous [−1, 1] m/s ²	Discrete(3)	Discrete(3)	Discrete(3)
Training budget	123 iters	2.5k / 9.36k iters	50k episodes	50k episodes

Result 1: Success & Collision Rate



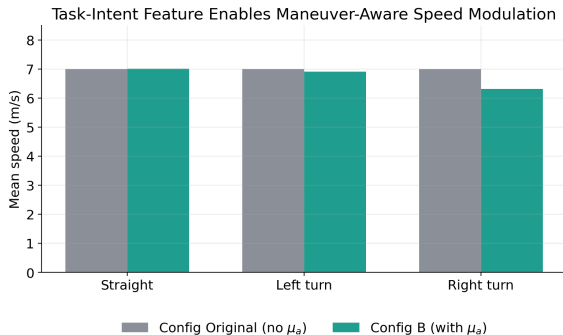
99.52% success, 0% collisions over 1,050 episodes — exceeds the paper's 97.4%/2.6%.

Result 2: Mean Crossing Speed



Nearly **2× faster** crossing than the paper — the simulators are not physically equivalent.

Result 3: Intent-Driven Speed Modulation



Config B ($+\mu_a$) slows for turns (Left 6.91, Right 6.32 m/s); Original stays uniform (~ 7.00 m/s).

Challenges and Limitations

Engineering & Reproduction (Resolved) ✓

Incomplete Paper Specification

Unreleased paper constants & parameters recovered via systematic inference.

Observation Space Alignment

Reconstructed unreleased 60/61-dim radar ordering, distance ranking, and presence masking.

Hybrid-CPU Hardware Bottleneck

Thread synchronization stalls resolved via Intel P-core affinity pinning (3.5× speedup).

Traditional RL Bottlenecks (Thesis Focus)

Non-Generalizable Baseline Policy

Current reproduction is a traditional baseline; lacks geometry invariance and cannot transfer.

Single Fixed Geometry Overfitting

Policy is overfit to a standard 4-way cross; fails on T-junctions, merges, or roundabouts.

Sim-to-Sim Kinematics Divergence

Micro-yielding behavior and collision physics differ between `highway-env` and paper sim.

Next Semester Focus: Generalizable Decision-Making

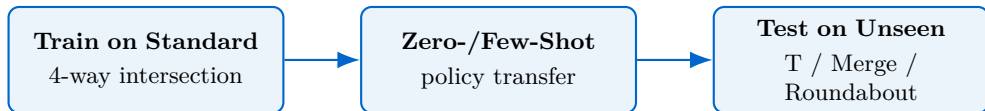
Core Limitation of Current Work

The reproduced Discrete SAC model is a **traditional single-geometry policy** tied to one 4-way layout. Real-world autonomous driving requires handling arbitrary junction topologies.

Thesis Objective (Next Semester): Develop a truly geometry-agnostic decision policy.

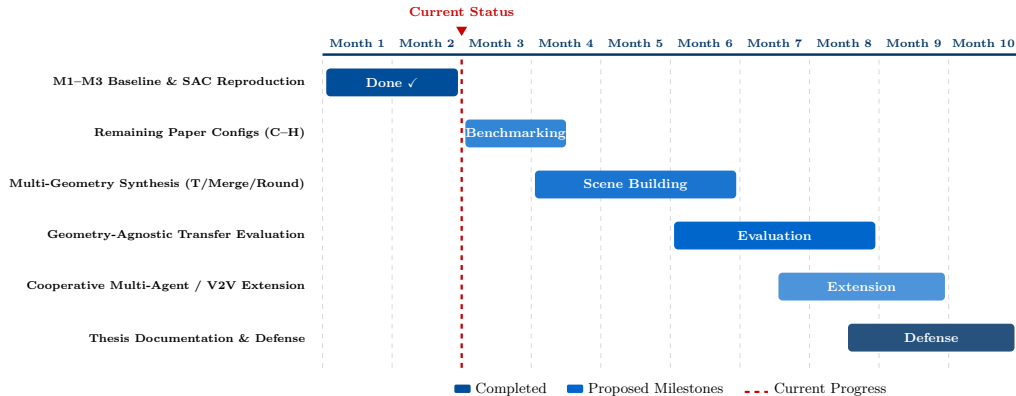
- **Invariant State Representation:** Replace fixed coordinates with conflict graphs and relative TTC
- **Cross-Topology Evaluation:** Zero-shot and few-shot transfer to T-junctions, roundabouts, and merges
- **Generalization Metric:** Quantify transfer of right-of-way negotiation without per-scenario retraining

Generalization Strategy



Graph-based and relational representations generalize far better across unseen junction layouts than fixed-coordinate vectors.

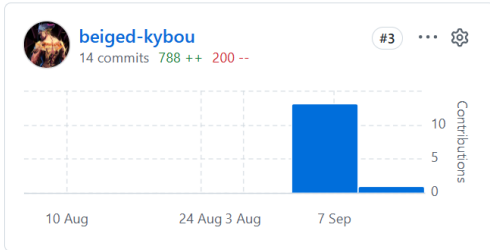
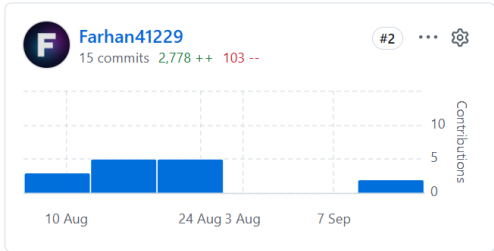
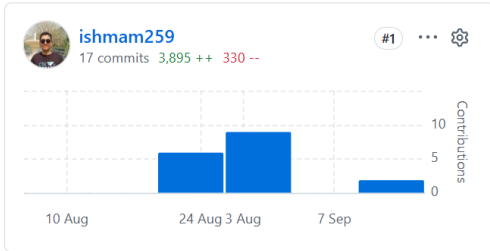
Proposed Project Timeline



Conclusion

- **Completed Milestone (Current Semester):** Rigorous 1:1 reproduction of the published **traditional RL baseline** (Discrete SAC, 99.52% success, 0% collisions over 1,050 tests).
- **Root-Cause Insights:** Identified simulator micro-yielding physics divergence and eliminated hybrid-CPU bottlenecks ($3.5\times$ throughput speedup).
- **Foundational Limitation:** Firmly established that traditional coordinate-based RL is **not generalizable** across intersection topologies.
- **Next Semester Thesis Research:** Formulated the concrete architecture and roadmap for **Generalizable, Geometry-Agnostic Decision-Making**.

GitHub Contributors Activity Graph



Thank You!

Questions & Discussion

Project Repository

<https://github.com/Farhan41229/Drl-unsignalized-intersection-navigation>

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